GapFlushRSS

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The process of performing a Dimensional Variation Analysis (DVA)—also known as tolerance stack-up analysis—on automotive interior parts involves a systematic, statistical approach to ensure components fit together precisely without unwanted interference, squeaks, or uneven gaps.
Here is the step-by-step process used by dimensional engineers to inform gap-setting decisions:

Architecture

- All deployed via docker to an lxc container on a proxmox server.
  LXC config required
  Docker config required

- Use python 3.13.12 as the base language to bring everything together. All code needs to be simple and understandable by a programmer who has only been coding for 6 months. Concepts need to be simple, well commented and clean.

Tools/Libraries

- Annotorious : A lightweight, open-source JavaScript library for image annotation. It supports custom shapes (bubbles/polygons), custom fields (numbers/text), and fires event listeners when a shape is created or clicked. Will require a custom bubble design which is then linked to the DB entry. The bubble will look like what is show in this image in RED. This will require customs CSS for Annotorious.



- The DB to use is PostgreSQL https://www.postgresql.org/
- Images should be stored using MinIO https://www.min.io/
- Use react for the website framework https://react.dev/
- Use Nginx as the http webserver https://nginx.org/
- Use openpyxl for exporting reports https://openpyxl.readthedocs.io/
- Use 0Auth for user management

Website UX

Simple clean UX switchable between a light theme and dark model
Colours to be simple Headings to be based around the colour green and it's shades.
Normal text to be black or inverted white when in dark mode.
Common header bar.
Use of pull down list for selecting projects and commodities.

Authorisation and User management

- Needs to work with Microsoft 0Auth maybe using keycloak for dev env.
- Microsoft Entra ID Developer Tenant for production.
- Needs a simple user name and password system as an alternative with user management and password recovery.
- Selectable between 0Auth and simple user name and password.

Users structure and user groups

Uses a hierarchical Role-Based Access Control (RBAC) system combined with project-scoped user groups.

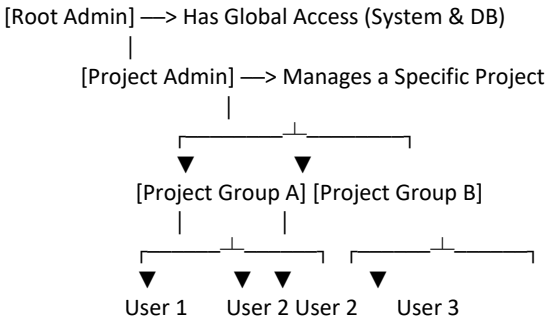
Role Hierarchy & Definitions

Define three global roles to establish system privileges:

* Root Admin (Global System Admin): Full access to global configuration, system settings, user creation, and database administration.
* Project Admin: Access to create new projects and manage assets/users within their assigned projects.
* Standard User: Access to view or interact with specific project data based on group assignments. [1]

Group-Based Scoping Architecture

To reduce administrative overhead, permissions are never assigned directly to individual users. Instead, implement a Scoped Group structure where groups are bound to specific projects. [2]



Project Group Templates

Create standard functional groups for every new project. When a Project Admin creates a new project (e.g., Project Alpha), the system automatically spins up a dedicated set of groups:

* Project-Alpha-Managers: Inherits read/write/delete permissions for Project Alpha.
* Project-Alpha-Contributors: Inherits read/write permissions for Project Alpha.
* Project-Alpha-Viewers: Inherits read-only permissions for Project Alpha.

The Group Mapping Table (Database Level)

To prevent creating an infinite number of global groups, use a join table in your database that links users, groups, and project scopes.

* Users Table: user_id, username, global_role (e.g., Root, Project Admin, User).
* Projects Table: project_id, project_name.
* Groups Table: group_id, group_name (e.g., Manager, Contributor, Viewer).
* User_Group_Project Map (The Core): user_id + group_id + project_id.

Operational Lifecycle (Reducing Overhead)

* Onboarding: When a new employee joins a team, the Project Admin simply adds their user_id to Project-Alpha-Contributors. They instantly inherit all relevant access.
* Offboarding/Transfer: If a user moves to a different team, removing them from the Project Alpha groups strips all access across that project's database entries and files in one click.
* Database Administration: The Root Admin uses a distinct, isolated global permission. Database migrations and configuration schemas should check for global_role == 'Root' before execution, bypassing project-level scope checks entirely. [3]

1. Define the Objectives and Gap Profiles

* Identify Critical Dimensions: Locate all critical-to-quality (CTQ) clearances.
* Set Nominal Targets: Establish the ideal engineering target dimensions (e.g., a nominal gap of 3.5 mm).

Gap and Flush planning page webpage. An image of the part or assembly will be uploaded to the site. The user will then mark up the image (using Annotorious) where the interfaces for analysis are. This will create an empty tolerance analysis (db entry) which will have a unique ID. The prefix ID will be a 2 letter code from the commodity list followed by a three number sequential code starting 001 (base number) will make up the rest of the unique ID.
Each Analysis can have either a gap analysis and/or a flush analysis attached but it must have one associated with it. Each one added will take the unique ID and add either -GAP or -FLUSH as a suffix. Each analysis of GAP or FLUSH will have a manually entered nominal target.

2. Establish the Geometric Alignment (Locating Schemes)

* Define the Principal Locating Profile (PLP): Determine how each part is physically constrained in 3D space using the 3-2-1 locating principle (primary, secondary, and tertiary locators like pins, holes, and slots).
* Establish the Measurement Coordinate System: Align all parts to the standard vehicle coordinate system (X: Fore/Aft, Y: Inboard/Outboard, Z: Up/Down).

The software will not need to provide anything for this it will be calculated by the user and will not be recorded. This maybe a module we add in the future.

3. Build the Geometric Model and Input Tolerances

* Map the Tolerance Chain: Trace the physical path of dimensions and interfaces from Component A, through the underlying structure, to Component B. This could theoretically be many parts.
* Input Component Tolerances: Apply manufacturing variations to the model based on Geometric Dimensioning.
* Account for Assembly Variation: Factor in the tolerances of the assembly process itself, including robotic placement variations, fixture tolerances, locator tolerances, fastening shifts, and other unique variations.

This is the key to the software. It should be possible to configure process types from a pull down such as plastics, castings, pressings, etc. This list should be editable by the admins.

The initial release will concentrate on plastics. Specifically DIN 16742 and the older standard DIN 16901.

When each parts material characteristics have been added and recorded we can move to the next step which is completing and RSS calculation for the objectives

4. Run the Statistical Simulation

* Choose Analysis Method: Use Root Sum Square (RSS) for simple linear stacks.

We are interested in 3 sigma, 4 sigma, and 6 sigma.
The distribution should also be shown on a graph view.

5. Evaluate Results and Inform Gap Decisions

The nominal target will be evaluated against the output of the Statistical simulation in stage 4

A visual RAG will be used help the user identify if the selected user nominal is compatible with the statistical simulation.

6.Report

*Each GAP and FLUSH analysis will require a printable report
*The report should also be exportable to excel via OpenPyXL



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